| Question | Answer | Mark | Mark scheme | Additional guidance |
|---|---|---|---|---|
| 28 | Complete chain of reasoning | M1 | for (area of trapezium $TQRS$ =) $0.5 \times 4x \times (2x + 3x)$<br>or for (area of rectangle $TUVS$ =) $4x \times (3x + 5)$ $(= 12x^2 + 20x)$ | Evidence for the award of marks may be seen on the diagram |
| | | M1 | for (area of trapezium $QUV\,R$ =) $4x(3x + 5) – 0.5 \times 4x \times (2x + 3x)$ | Alternative methods may be seen. |
| | | C1 | for correct algebraic processing and simplification to the given form | |
| | | M1 | **Alternative 1**<br>for ($QU$ =) $3x + 5 – 2x$ $(= x + 5)$ | |
| | | M1 | for (area of trapezium $QUVR$ =) $0.5 \times 4x \times$ (("$x + 5$") + 5) or $0.5 \times 4x \times (x + 10)$ | |
| | | C1 | for correct algebraic processing and simplification to the given form | |
| | | M1 | **Alternative 2**<br>for (area of triangle =) $0.5 \times (3x – 2x) \times 4x$ or for (area of rectangle =) $4x \times 5$ | Accept $x$ for $(3x – 2x)$ |
| | | M1 | for (area of trapezium $QUVR$ =) "$0.5 \times (3x – 2x) \times 4x$" + "$4x \times 5$" | |
| | | C1 | for correct algebraic processing and simplification to the given form | |